

Zero-Shot, One-Shot, & Few-Shot Prompting: These are prompting techniques where a model is given zero, one, or a few examples of a task to guide its response. Providing more examples generally helps the model better understand the user's intent and improves its accuracy for the specific task.

Multimodality: Multimodality is an AI's ability to understand and process information across multiple data types like text, images, and audio. This allows for more versatile and human-like interactions, such as describing an image or answering a spoken question.

Grounding: Grounding is the process of connecting a model's outputs to verifiable, real-world information sources to ensure factual accuracy and reduce hallucinations. This is often achieved with techniques like RAG to make AI systems more trustworthy.

Core AI Model Architectures

Transformers: The Transformer is the foundational neural network architecture for most modern LLMs. Its key innovation is the self-attention mechanism, which efficiently processes long sequences of text and captures complex relationships between words.

Recurrent Neural Network (RNN): The Recurrent Neural Network is a foundational architecture that preceded the Transformer. RNNs process information sequentially, using loops to maintain a "memory" of previous inputs, which made them suitable for tasks like text and speech processing.

Mixture of Experts (MoE): Mixture of Experts is an efficient model architecture where a "router" network dynamically selects a small subset of "expert" networks to handle any given input. This allows models to have a massive number of parameters while keeping computational costs manageable.

Diffusion Models: Diffusion models are generative models that excel at creating high-quality images. They work by adding random noise to data and then training a model to meticulously reverse the process, allowing them to generate novel data from a random starting point.

Mamba: Mamba is a recent AI architecture using a Selective State Space Model (SSM) to process sequences with high efficiency, especially for very long contexts. Its selective mechanism allows it to focus on relevant information while filtering out noise, making it a potential alternative to the Transformer.

The LLM Development Lifecycle

The development of a powerful language model follows a distinct sequence. It begins with Pre-training, where a massive base model is built by training it on a vast dataset of general internet text to learn language, reasoning, and world knowledge. Next is Fine-tuning, a specialization phase where the general model is further trained on smaller, task-specific datasets to adapt its capabilities for a particular purpose. The final stage is Alignment, where the specialized model's behavior is adjusted to ensure its outputs are helpful, harmless, and aligned with human values.

Pre-training Techniques: Pre-training is the initial phase where a model learns general knowledge from vast amounts of data. The top techniques for this involve different objectives for the model to learn from. The most common is Causal Language Modeling (CLM), where the model predicts the next word in a sentence. Another is Masked Language Modeling (MLM), where the model fills in intentionally hidden words in a text. Other important methods include Denoising Objectives, where the model learns to restore a corrupted input to its original state, Contrastive Learning, where it learns to distinguish between similar and dissimilar pieces of data, and Next Sentence Prediction

(NSP), where it determines if two sentences logically follow each other.

Fine-tuning Techniques: Fine-tuning is the process of adapting a general pre-trained model to a specific task using a smaller, specialized dataset. The most common approach is Supervised Fine-Tuning (SFT), where the model is trained on labeled examples of correct input-output pairs. A popular variant is Instruction Tuning, which focuses on training the model to better follow user commands. To make this process more efficient, Parameter-Efficient Fine-Tuning (PEFT) methods are used, with top techniques including LoRA (Low-Rank Adaptation), which only updates a small number of parameters, and its memory-optimized version, QLoRA. Another technique,

Retrieval-Augmented Generation (RAG), enhances the model by connecting it to an external knowledge source during the fine-tuning or inference stage.

Alignment & Safety Techniques:

Alignment is the process of ensuring an AI model's behavior aligns with human values and expectations, making it helpful and harmless. The most prominent technique is Reinforcement Learning from Human Feedback (RLHF), where a "reward model" trained on human preferences guides the AI's learning process, often using an algorithm like Proximal Policy Optimization (PPO) for stability. Simpler alternatives have emerged, such as Direct Preference Optimization (DPO), which bypasses the need for a separate reward model, and Kahneman-Tversky Optimization (KTO), which simplifies data collection further. To ensure safe

deployment, Guardrails are implemented as a final safety layer to filter outputs and block harmful actions in real-time.

Enhancing AI Agent Capabilities

AI agents are systems that can perceive their environment and take autonomous actions to achieve goals. Their effectiveness is enhanced by robust reasoning frameworks.

Chain of Thought (CoT): This prompting technique encourages a model to explain its reasoning step-by-step before giving a final answer. This process of "thinking out loud" often leads to more accurate results on complex reasoning tasks.

Tree of Thoughts (ToT): Tree of Thoughts is an advanced reasoning framework where an agent explores multiple reasoning paths simultaneously, like branches on a tree. It allows the agent to self-evaluate different lines of thought and choose the most promising one to pursue, making it more effective at complex problem-solving.

ReAct (Reason and Act): ReAct is an agent framework that combines reasoning and acting in a loop. The agent first "thinks" about what to do, then takes an "action" using a tool, and uses the resulting observation to inform its next thought, making it highly effective at solving complex tasks.

Planning: This is an agent's ability to break down a high-level goal into a sequence of smaller, manageable sub-tasks. The agent then creates a plan to execute these steps in order, allowing it to handle complex, multi-step assignments.

Deep Research: Deep research refers to an agent's capability to autonomously explore a topic in-depth by iteratively searching for information, synthesizing findings, and identifying new questions. This allows the agent to build a comprehensive understanding of a subject far beyond a single search query.

Critique Model: A critique model is a specialized AI model trained to review, evaluate, and provide feedback on the output of another AI model. It acts as an automated critic, helping to identify errors, improve reasoning, and ensure the final output meets a desired quality standard.

Index of Terms

This index of terms was generated using Gemini Pro 2.5. The prompt and reasoning steps are included at the end to demonstrate the time-saving benefits and for educational purposes.

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- A/B Testing - Chapter 3: Parallelization
- Action Selection - Chapter 20: Prioritization
- Adaptation - Chapter 9: Learning and Adaptation
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Prompt

Below is the prompt that was utilized to create this index of terms.

*do you read
<https://docs.google.com/document/d/1rsaK53T3Lg5KoGwvf8ukOUvbELRtH-V0LnOIFDxBryE/edit?tab=t.0> and all the related documents in the first page
Please extract all the titles for each chapter. and print them.
Then process the content of all the individual chapters, and create a detailed index used for publishing, a list of words or phrases ('headings') and associated pointers ('locators') to where useful material relating to that heading can be found in a document or collection of documents. I don't want a sample index. I want the full index, with 100 index terms*

Gemini 2.5: Reasoning Steps

Beginning Index Creation

I've initiated the process by reading the Google Doc provided. Currently, I'm focusing on understanding the user's request: extracting chapter titles to build a detailed index. I'm carefully reviewing the document, looking for clear structural elements that would indicate chapter breaks. I'm hoping to identify consistent formatting or keywords that will guide the indexing process.

Defining Index Extraction Process

I'm now fully immersed in the task. I've broken down the user's request into actionable steps. My initial focus is using Workspaces to access the main Google Doc. Following this, I'll parse the document to locate and retrieve related documents, which will also be accessed using the same tool. Finally, I will extract all chapter titles from all the documents, and create a comprehensive index. I've planned the detailed process.

Refining the Extraction Strategy

The user's instructions are clear: read a Google Doc, its related documents, and build a detailed index of chapter titles. I've devised a multi-step plan. First, Workspaces will access the main Google Doc. Then, I'll parse it to find and read related documents, also using Workspaces. I'm now focusing on extracting titles and forming the detailed index with the required terms. I'm prioritizing accuracy to meet the user's specific demands for a comprehensive index.

1 . Establish user intent:

I've determined the user wants a chapter title list and comprehensive index from the Google Doc and related documents. Now I need to figure out the most effective method for this extraction and indexing process. I'm exploring potential API functionalities to streamline this task, and will need to assess the best strategy for handling different document structures.

3. Considering Tool Use

I've extracted the chapter URLs from the main document using a regular expression. Now, I'm ready to utilize Workspaces to retrieve the actual content of each chapter. Once I have the text for each chapter, I'll proceed with indexing.

Accessing chapter content:

I've successfully identified the URLs for each chapter from the main document, which I can now use with Workspaces. My next step is to retrieve the content from these URLs, which will then allow me to generate the index. I'm focusing on efficiency, aiming to process this information quickly.

Fetching Chapter Content

I've successfully identified the chapter URLs from the main document and am ready to utilize Workspaces. My plan involves extracting these URLs, batching the calls for efficiency, and then processing the retrieved content to create the desired index. This ensures all relevant information is captured, paving the way for a comprehensive and accurate result.

Initiating URL Retrieval

I've now successfully extracted the chapter URLs from the main document. Currently, I'm setting up calls to the Workspaces tool, preparing to fetch the content of each chapter. The goal is to obtain the necessary text for indexing. This information will then facilitate the creation of the index with at least 100 terms as requested.

Initiating Chapter Extraction

I've successfully identified and extracted the chapter URLs from the main document, which I've included above. I am now proceeding to use Workspaces for each URL to obtain the chapter content. The goal is to obtain the necessary text for indexing. This information will then facilitate the creation of the index with at least 100 terms as requested. I've begun to retrieve this information.

Orchestrating Chapter Retrieval

I've extracted the chapter URLs and initiated a call to Workspaces. I'm currently retrieving the content for each chapter. To ensure a smooth process, I'm batching the calls to prevent any rate-limiting issues. This should enable me to efficiently compile the text needed for the index.

Decoding User Request

I've determined the user's need. They require two primary deliverables: first, a comprehensive list of chapter titles extracted from Google Docs, and second, a detailed index containing a minimum of one hundred key terms. The process of extracting titles seems straightforward, while curating the index will involve more nuanced analysis.

Extracting Titles and Indexing

I've successfully identified the user's need: extracting chapter titles from the provided documents and compiling a comprehensive index. I've fetched the content of 31 Google Docs, including the main document and its chapters. I've confirmed that no additional tools are needed. My plan involves two steps: first, extracting chapter titles from the document filenames, and second, meticulously processing the text to build the index with key terms and chapter locators.

Developing an Extraction Plan

I've refined my approach to tackle the user's request. I've now formulated a precise plan. Firstly, I will compile the chapter titles from the filenames of the 31 fetched documents. Secondly, and more complexly, I will meticulously analyze the content to create a comprehensive, alphabetically ordered index exceeding 100 entries. I have finalized my extraction and indexing strategy and can now commence execution.

Crafting the Extraction Strategy

My approach is well-defined. I will compile chapter titles from the filenames of the 31 fetched documents. Subsequently, I'll analyze the content meticulously to construct an alphabetically organized index exceeding 100 entries. I've finalized the extraction and indexing strategy and can now commence implementation. The user's goal of a detailed index, covering content from diverse chapters and appendices within the Agentic Design Patterns series, is now within reach. I'll focus on delivering a thorough response, going beyond a mere sample index.

Defining the Approach

I've clarified the user's need. I will extract chapter titles from the document filenames. Following that, I will carefully create an index exceeding 100 entries, arranging them

alphabetically and noting their chapter locations. I am prepared to begin the indexing and extraction process.

Frequently Asked Questions: Agentic Design Patterns

What is an "agentic design pattern"? An agentic design pattern is a reusable, high-level solution to a common problem encountered when building intelligent, autonomous systems (agents). These patterns provide a structured framework for designing agent behaviors, much like software design patterns do for traditional programming. They help developers build more robust, predictable, and effective AI agents.

What is the main goal of this guide? The guide aims to provide a practical, hands-on introduction to designing and building agentic systems. It moves beyond theoretical discussions to offer concrete architectural blueprints that developers can use to create agents capable of complex, goal-oriented behavior in a reliable way.

Who is the intended audience for this guide? This guide is written for AI developers, software engineers, and system architects who are building applications with large language models (LLMs) and other AI components. It is for those who want to move from simple prompt-response interactions to creating sophisticated, autonomous agents.

4. What are some of the key agentic patterns discussed? Based on the table of contents, the guide covers several key patterns, including:

- **Reflection:** The ability of an agent to critique its own actions and outputs to improve performance.
- **Planning:** The process of breaking down a complex goal into smaller, manageable steps or tasks.
- **Tool Use:** The pattern of an agent utilizing external tools (like code interpreters, search engines, or other APIs) to acquire information or perform actions it cannot do on its own.
- **Multi-Agent Collaboration:** The architecture for having multiple specialized agents work together to solve a problem, often involving a "leader" or "orchestrator" agent.
- **Human-in-the-Loop:** The integration of human oversight and intervention, allowing for feedback, correction, and approval of an agent's actions.

Why is "planning" an important pattern? Planning is crucial because it allows an agent to tackle complex, multi-step tasks that cannot be solved with a single action. By creating a plan, the agent can maintain a coherent strategy, track its progress, and handle errors or unexpected obstacles in a structured manner. This prevents the agent from getting "stuck" or deviating from the user's ultimate goal.

What is the difference between a "tool" and a "skill" for an agent? While the terms are often used interchangeably, a "tool" generally refers to an external resource the agent can call upon (e.g., a weather API, a calculator). A "skill" is a more integrated capability that the agent has learned, often combining tool use with internal reasoning to perform a specific function (e.g., the skill of "booking a flight" might involve using calendar and airline APIs).

How does the "Reflection" pattern improve an agent's performance? Reflection acts as a form of self-correction. After generating a response or completing a task, the agent can be prompted to review its work, check for errors, assess its quality against certain criteria, or consider alternative approaches. This iterative refinement process helps the agent produce more accurate, relevant, and high-quality results.

What is the core idea of the Reflection pattern? The Reflection pattern gives an agent the ability to step back and critique its own work. Instead of producing a final output in one go, the agent generates a draft and then "reflects" on it, identifying flaws, missing information, or areas for improvement. This self-correction process is key to enhancing the quality and accuracy of its responses.

Why is simple "prompt chaining" not enough for high-quality output? Simple prompt chaining (where the output of one prompt becomes the input for the next) is often too basic. The model might just rephrase its previous output without genuinely improving it. A true Reflection pattern requires a more structured critique, prompting the agent to analyze its work against specific standards, check for logical errors, or verify facts.

What are the two main types of reflection mentioned in this chapter? The chapter discusses two primary forms of reflection:

- **"Check your work" Reflection:** This is a basic form where the agent is simply asked to review and fix its previous output. It's a good starting point for catching simple errors.
- **"Internal Critic" Reflection:** This is a more advanced form where a separate, "critic" agent (or a dedicated prompt) is used to evaluate the output of the "worker" agent. This critic can be given specific criteria to look for, leading to more rigorous and targeted improvements.

How does reflection help in reducing "hallucinations"? By prompting an agent to review its work, especially by comparing its statements against a known source or by checking its own reasoning steps, the Reflection pattern can significantly reduce the likelihood of hallucinations (making up facts). The agent is forced to be more grounded in the provided context and less likely to generate unsupported information.

Can the Reflection pattern be applied more than once? Yes, reflection can be an iterative process. An agent can be made to reflect on its work multiple times, with each loop refining the output further. This is particularly useful for complex tasks where the first or second attempt may still contain subtle errors or could be substantially improved.

What is the Planning pattern in the context of AI agents? The Planning pattern involves enabling an agent to break down a complex, high-level goal into a sequence of smaller, actionable steps. Instead of trying to solve a big problem at once, the agent first creates a "plan" and then executes each step in the plan, which is a much more reliable approach.

Why is planning necessary for complex tasks? LLMs can struggle with tasks that require multiple steps or dependencies. Without a plan, an agent might lose track of the overall

objective, miss crucial steps, or fail to handle the output of one step as the input for the next. A plan provides a clear roadmap, ensuring all requirements of the original request are met in a logical order.

What is a common way to implement the Planning pattern? A common implementation is to have the agent first generate a list of steps in a structured format (like a JSON array or a numbered list). The system can then iterate through this list, executing each step one by one and feeding the result back to the agent to inform the next action.

How does the agent handle errors or changes during execution? A robust planning pattern allows for dynamic adjustments. If a step fails or the situation changes, the agent can be prompted to "re-plan" from the current state. It can analyze the error, modify the remaining steps, or even add new ones to overcome the obstacle.

Does the user see the plan? This is a design choice. In many cases, showing the plan to the user first for approval is a great practice. This aligns with the "Human-in-the-Loop" pattern, giving the user transparency and control over the agent's proposed actions before they are executed.

What does the "Tool Use" pattern entail? The Tool Use pattern allows an agent to extend its capabilities by interacting with external software or APIs. Since an LLM's knowledge is static and it can't perform real-world actions on its own, tools give it access to live information (e.g., Google Search), proprietary data (e.g., a company's database), or the ability to perform actions (e.g., send an email, book a meeting).

How does an agent decide which tool to use? The agent is typically given a list of available tools along with descriptions of what each tool does and what parameters it requires. When faced with a request it can't handle with its internal knowledge, the agent's reasoning ability allows it to select the most appropriate tool from the list to accomplish the task.

What is the "ReAct" (Reason and Act) framework mentioned in this context? ReAct is a popular framework that integrates reasoning and acting. The agent follows a loop of **Thought** (reasoning about what it needs to do), **Action** (deciding which tool to use and with what inputs), and **Observation** (seeing the result from the tool). This loop continues until it has gathered enough information to fulfill the user's request.

What are some challenges in implementing tool use? Key challenges include:

- **Error Handling:** Tools can fail, return unexpected data, or time out. The agent needs to be able to recognize these errors and decide whether to try again, use a different tool, or ask the user for help.
- **Security:** Giving an agent access to tools, especially those that perform actions, has security implications. It's crucial to have safeguards, permissions, and often human approval for sensitive operations.
- **Prompting:** The agent must be prompted effectively to generate correctly formatted tool calls (e.g., the right function name and parameters).

What is the Human-in-the-Loop (HITL) pattern? HITL is a pattern that integrates human oversight and interaction into the agent's workflow. Instead of being fully autonomous, the agent pauses at critical junctures to ask for human feedback, approval, clarification, or direction.

Why is HITL important for agentic systems? It's crucial for several reasons:

- **Safety and Control:** For high-stakes tasks (e.g., financial transactions, sending official communications), HITL ensures a human verifies the agent's proposed actions before they are executed.
- **Improving Quality:** Humans can provide corrections or nuanced feedback that the agent can use to improve its performance, especially in subjective or ambiguous tasks.
- **Building Trust:** Users are more likely to trust and adopt an AI system that they can guide and supervise.

At what points in a workflow should you include a human? Common points for human intervention include:

- **Plan Approval:** Before executing a multi-step plan.
- **Tool Use Confirmation:** Before using a tool that has real-world consequences or costs money.
- **Ambiguity Resolution:** When the agent is unsure how to proceed or needs more information from the user.
- **Final Output Review:** Before delivering the final result to the end-user or system.

Isn't constant human intervention inefficient? It can be, which is why the key is to find the right balance. HITL should be implemented at critical checkpoints, not for every single action. The goal is to build a collaborative partnership between the human and the agent, where the agent handles the bulk of the work and the human provides strategic guidance.

What is the Multi-Agent Collaboration pattern? This pattern involves creating a system composed of multiple specialized agents that work together to achieve a common goal. Instead of one "generalist" agent trying to do everything, you create a team of "specialist" agents, each with a specific role or expertise.

What are the benefits of a multi-agent system?

- **Modularity and Specialization:** Each agent can be fine-tuned and prompted for its specific task (e.g., a "researcher" agent, a "writer" agent, a "code" agent), leading to higher quality results.
- **Reduced Complexity:** Breaking a complex workflow down into specialized roles makes the overall system easier to design, debug, and maintain.
- **Simulated Brainstorming:** Different agents can offer different perspectives on a problem, leading to more creative and robust solutions, similar to how a human team works.

What is a common architecture for multi-agent systems? A common architecture involves an **Orchestrator Agent** (sometimes called a "manager" or "conductor"). The orchestrator understands the overall goal, breaks it down, and delegates sub-tasks to the appropriate specialist agents. It then collects the results from the specialists and synthesizes them into a final output.

How do the agents communicate with each other? Communication is often managed by the orchestrator. For example, the orchestrator might pass the output of the "researcher" agent to the "writer" agent as context. A shared "scratchpad" or message bus where agents can post their findings is another common communication method.

Why is evaluating an agent more difficult than evaluating a traditional software program? Traditional software has deterministic outputs (the same input always produces the same output). Agents, especially those using LLMs, are non-deterministic and their performance can be subjective. Evaluating them requires assessing the *quality* and *relevance* of their output, not just whether it's technically "correct."

What are some common methods for evaluating agent performance? The guide suggests a few methods:

- **Outcome-based Evaluation:** Did the agent successfully achieve the final goal? For example, if the task was "book a flight," was a flight actually booked correctly? This is the most important measure.
- **Process-based Evaluation:** Was the agent's *process* efficient and logical? Did it use the right tools? Did it follow a sensible plan? This helps debug why an agent might be failing.
- **Human Evaluation:** Having humans score the agent's performance on a scale (e.g., 1-5) based on criteria like helpfulness, accuracy, and coherence. This is crucial for user-facing applications.

What is an "agent trajectory"? An agent trajectory is the complete log of an agent's steps while performing a task. It includes all its thoughts, actions (tool calls), and observations. Analyzing these trajectories is a key part of debugging and understanding agent behavior.

How can you create reliable tests for a non-deterministic system? While you can't guarantee the exact wording of an agent's output, you can create tests that check for key elements. For example, you can write a test that verifies if the agent's final response *contains* specific information or if it successfully called a certain tool with the right parameters. This is often done using mock tools in a dedicated testing environment.

How is prompting an agent different from a simple ChatGPT prompt? Prompting an agent involves creating a detailed "system prompt" or constitution that acts as its operating instructions. This goes beyond a single user query; it defines the agent's role, its available tools, the patterns it should follow (like ReAct or Planning), its constraints, and its personality.

What are the key components of a good system prompt for an agent? A strong system prompt typically includes:

- **Role and Goal:** Clearly define who the agent is and what its primary purpose is.
- **Tool Definitions:** A list of available tools, their descriptions, and how to use them (e.g., in a specific function-calling format).
- **Constraints and Rules:** Explicit instructions on what the agent *should not* do (e.g., "Do not use tools without approval," "Do not provide financial advice").
- **Process Instructions:** Guidance on which patterns to use. For example, "First, create a plan. Then, execute the plan step-by-step."
- **Example Trajectories:** Providing a few examples of successful "thought-action-observation" loops can significantly improve the agent's reliability.

What is "prompt leakage"? Prompt leakage occurs when parts of the system prompt (like tool definitions or internal instructions) are inadvertently revealed in the agent's final response to the user. This can be confusing for the user and expose underlying implementation details. Techniques like using separate prompts for reasoning and for generating the final answer can help prevent this.

What are some future trends in agentic systems? The guide points towards a future with:

- **More Autonomous Agents:** Agents that require less human intervention and can learn and adapt on their own.
- **Highly Specialized Agents:** An ecosystem of agents that can be hired or subscribed to for specific tasks (e.g., a travel agent, a research agent).
- **Better Tools and Platforms:** The development of more sophisticated frameworks and platforms that make it easier to build, test, and deploy robust multi-agent systems.